

Direction Sense

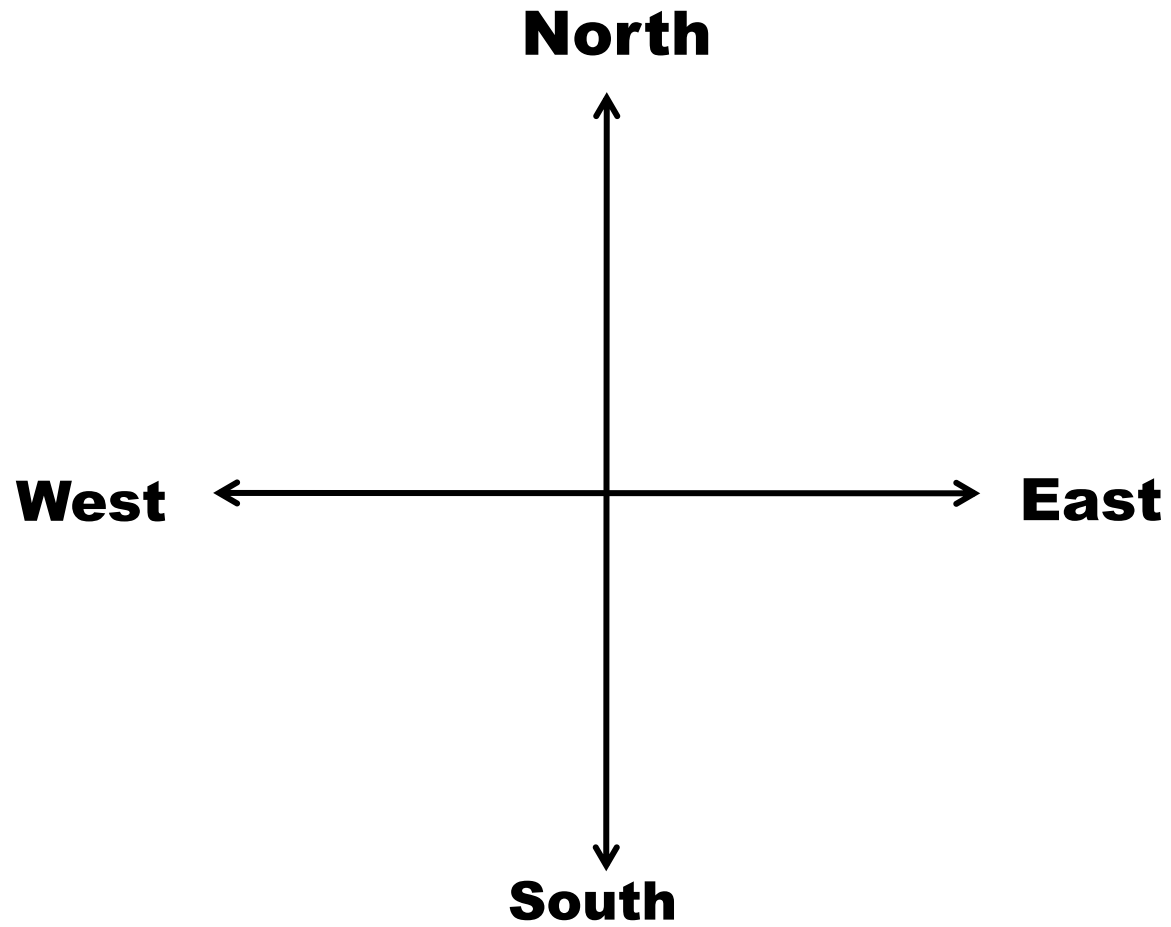


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Learning Objective

- Diagrammatic representation of the directions & the corresponding left and right
- The importance of scale while solving the questions
- The difference between total distance and minimum distance between two points
- Meaning and usage of Pythagorean triplets
- Major triplets like 3, 4, 5; 5, 12, 13; 7, 24, 25; 8, 15, 17.
- Simple problems to understand the concept





1. Raman walks 2 km towards North and turns to his right and walks 4 km. He turns right and walks 3 km and again turns right and walks 4 km. He now turns back. In which direction he is going now?

1. East
2. North
3. West
4. South

(Cncp : Directions)



2. Rahul walks 30 metres towards south. Then turns to his right and starts walking straight till he completes another 30 metres. Then again turning to his left he walks for 20 metres. He then turns to his left and walks for 30 metres. How far is he from his initial position?

1. 50 metres
2. 30 metres
3. 10 metres
4. 60 metres

(Cncp : Directions)



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3. Navin walked 20 metres towards East, took a left turn and walked 20 metres. How far is he from his starting position?

1. $20\sqrt{2}$ metres
2. $40\sqrt{2}$ metres
3. $50\sqrt{2}$ metres
4. $30\sqrt{2}$ metres

(Cncp : Directions)



4. Reeta drives to North of her place of stay at A and finds after travelling 25 km that she has driven in the wrong direction. She then turns to the right and travels 2 km and then again turns right and drives straight another 25 km. How much distance she has now to cover to go back to the point from where she started?

1. 25 km
2. 4 km
3. 2 km
4. 50 km

(Cncp : Directions)



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5. A mountaineer starts from Camp A and proceeds towards East to Camp B 4 Km away. From Camp B, he proceeds to Camp C, 5 km to the south. From there he returns and proceeds 12 km to Camp D via Camps B and A. How far is he from the Camp A?

(Cncp : Directions)



6. Boys and Girls are made to sit in 5 rows and 5 columns all of them facing North. A is at the center of the arrangement, P is to the immediate right of A, Q is just behind P and R is north of A, Q is in which direction with respect to R?

(Cncp : Directions)



7. Ram walks east for 15 km and then turns right and walks for 10 km and then turns left and walks for 20 km, again he turns left and walks for 40 Km. His direction now w.r.t the position he started is

(Cncp : Directions)



8. Rajeev traveled 28 km towards East from a point 'S' and then turning towards right traveled 21 km and reached a point 'F' . How far is he from the starting point (Shortest distance in km) and in which direction?

(Cncp : Directions)



Class Exercise – I from PDF



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9. A boy runs 20 m towards east and then turns to right, walks 10 m, again turns to right, runs 9 m, then moves to 5 m and 12 m to his left each time and then finally moves 6 m by turning to his right. In which direction is the boy facing in the end?

(Cncp : Directions)



10. There is a ring road connecting points A, B, C and E. The road is in a complete circular form but having several approach roads leading to the centre. Exactly in the centre of the ring road there is a tree which is 20 km from point A on the circular road. You have taken a round of the circular road starting from point A and finish at the same point after touching points B, C and E. You then drive 20 km interior towards the tree from point A and from there reach somewhere in between B and C on the ring road. How much distance you have to travel from the tree to reach the point between B and C on the ring road?

1. 20 km

2. 15 km

3. 80 km

4. 40 km

(Cncp : Directions)



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Q. There is a ring road connecting points A, B, C and E. The road is in a complete circular form but having several approach roads leading to the centre. Exactly in the centre of the ring road there is a tree which is 20 km from point A on the circular road. You have taken a round of the circular road starting from point A and finish at the same point after touching points B, C and E. You then drive 20 km interior towards the tree from point A and from there reach somewhere in between B and C on the ring road. How much distance you have to travel from the tree to reach the point between B and C on the ring road?

1. 20 km
2. 15 km
3. 80 km
4. 40 km



11. A watch reads 4.30. If the minute hand points East, in what direction will the hour hand point?

1. North
2. North-west
3. North-east
4. South-east

(Cncp : Directions)



12. Bhavika and Sunaina start simultaneously towards each other from places 100 m apart. After walking 30 m Bhavika turns left and goes 10 m, then she turns right and goes 20 m and then turns right again and comes back to the road on which she had started walking. If both Bhavika and Sunaina walk with the same speed, what is the distance between them at this point of time?

1. 70 metres
2. 10 metres
3. 40 metres
4. 20 metres

(Cncp : Directions)



hitbullseye *Test Prep Simplified....*

Q. Bhavika and Sunaina start simultaneously towards each other from places 100 m apart. After walking 30 m Bhavika turns left and goes 10 m, then she turns right and goes 20 m and then turns right again and comes back to the road on which she had started walking. If both Bhavika and Sunaina walk with the same speed, what is the distance between them at this point of time?

1. 70 metres
2. 10 metres
3. 40 metres
4. 20 metres



13. The door of Aditya's house faces East. From the back side of his house, he walks straight 50 metres, then turns to the right and walks 50 metres again. Finally, he turns towards left and stops after walking 25 metres. Now, Aditya is in which direction from the starting point?

1. South-East
2. South-West
3. North-East
4. North-West

(Cncp : Directions)



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14. A man starts from a point and moves 3 km north, then turns to west and goes 2 km. He turns north and walks 1km and then moves 5 km towards east. How far is he from the starting point by following shortest path?

1. 11 km
2. 10 km
3. 5 km
4. 8 km

(Cncp : Directions)



15. Rajesh walked 25 meters towards South. Then he turned to his left and walked 20 meters. He then turned to his left and walked 25 meters. He again turned to his right and walked 15 meters. At what distance is he from the starting point and in which direction?

1. 60 meters – East
2. 35 meters – East
3. 35 meters – North
4. 40 meters – East

(Cncp : Directions)



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Q. Rajesh walked 25 meters towards South. Then he turned to his left and walked 20 meters. He then turned to his left and walked 25 meters. He again turned to his right and walked 15 meters. At what distance is he from the starting point and in which direction?

1. 60 meters – East
2. 35 meters – East
3. 35 meters – North
4. 40 meters – East



Class Exercise – II from PDF



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16. A person goes 30 km North, then turns right and walks 40 km, then again turns right and walks 20 km then again turns right and walks 40 km. How many kilometres is he from his original position?

1. 5
2. 10
3. 15
4. 20

(Cncp : Directions)



17. Shaloo ran 20m to the east, then he turned left and walked for 15 m, then turned right and went 25 m and then turned right again and went 15 m. How far was Shaloo from the starting point?

1. 45m
2. 35m
3. 25m
4. 15m

(Cncp : Directions)



DIRECTIONS for questions: *Study the information and answer the questions.*

1. There are 6 check-posts, A, B, C, D, E and F.
2. Check-post D is 15 kms to the North of F which is 25 kms to the North- East of B.
3. Check-post A is 5 kms west of E and 15 kms to the South-West of 'C'.
4. B, A and E are in straight line in the same order.
5. The check posts B and E are 30 kms apart from each other.

(Cncp : Directions)



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5. The check posts B and E are 30 kms apart from each other.

18. Which check post is the farthest to the South-West of D?

1. A
2. B
3. C
4. D



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3. Check-post A is 5 kms west of E and 15 kms to the South-West of 'C'.
4. B, A and E are in straight line in the same order.
5. The check posts B and E are 30 kms apart from each other.

19. Which post is the nearest and to the North-East of E?

1. A
2. B
3. C
4. D



1. There are 6 check-posts, A, B, C, D, E and F.
2. Check-post D is 15 kms to the North of F which is 25 kms to the North-East of B.
3. Check-post A is 5 kms west of E and 15 kms to the South-West of 'C'.
4. B, A and E are in straight line in the same order.
5. The check posts B and E are 30 kms apart from each other.

20. If a jeep moves from E to D via A, B and F, how much distance it will have to cover?

1. 70 km
2. 120 km
3. 100 km
4. 80 km





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Answer key

Q. no	Key	Q. no	Key
1.	1 st option	13.	4 th option
2.	1 st option	14.	3 rd option
3.	1 st option	15.	2 nd option
4.	3 rd option	16.	2 nd option
5.	3 km	17.	1 st option
6.	South – East	18.	2 nd option
7.	North – East	19.	3 rd option
8.	35 km, South – East	20.	1 st option
9.	South		
10.	1 st option		
11.	3 rd option		
12.	4 th option		

Explanation

Q. No.	Ans.	Explanations
1.	1	As shown he is going towards east.
2.	1	So he is 50m for from his initial position

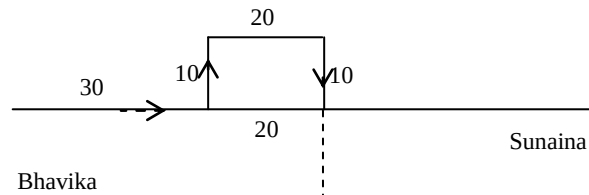
3.	1	$OB^2 = OA^2 + AB^2$ $= 20^2 + 20^2$ $= 400 + 400 = 800$ $OB = \sqrt{800} = 20\sqrt{2}$
4.	3	As shown it is 2 km.

5.	3 km	As shown D is 12 km from C.
6.	South-East	

		As shown Q is South-East of R.
7.	North-East	As shown he is in north-East.

12.

4

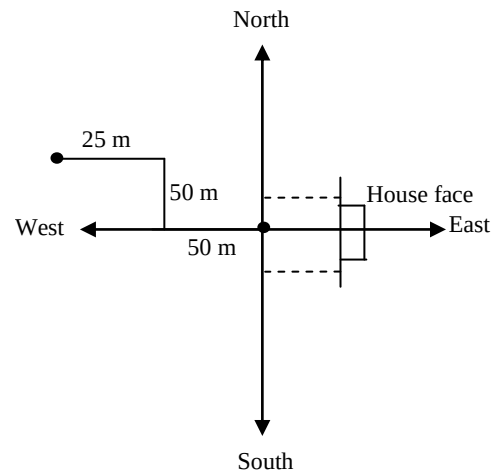


Total distance travelled by Bhavika is 70m,(but on actual route she has covered 50 m). So is by sunaina (as both have same speed)
So distance between them = $70 - 50 = 20\text{m}$.

13.

4

As shown it is North West.



14.

3

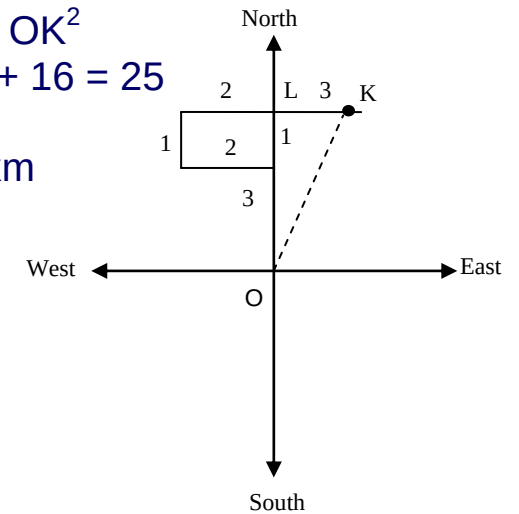
$$OL^2 + LK^2 = OK^2$$

$$4^2 + 3^2 = OK^2$$

$$OK^2 = 9 + 16 = 25$$

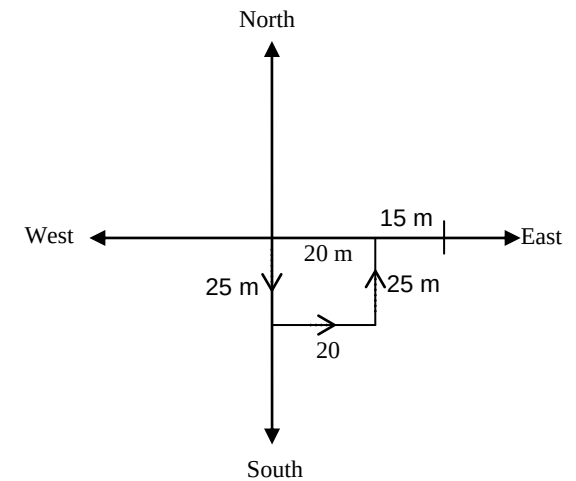
$$OK^2 = 5^2$$

$$OK = 5 \text{ km}$$



15.

2



The distanc is 35 m East.

